Diversified Ecommerce Affordability and Service Risk Analytics

SQL Phase 2 Capstone Project - Prepared for Shraddha Yadav

1. Domain, Dataset and End Consumer

Domain: Employment & Poverty / Consumer Access - ecommerce pricing, availability and service risk

Dataset: diversified_ecommerce_dataset.csv

Cleaned SQLite table name: diversified_ecommerce

Rows used in project database: 5,000. Columns: 21.

End consumer: A retail operations and customer-access analyst who needs to understand affordability, product performance, customer reach, return risk, and operational planning issues from transaction/product data.

Important approval note: this pack is approval-first. Send the dataset approval email before treating the SQL/report as final submission.

2. Five Analytical Perspectives

1. Affordability and Pricing
2. Inventory and Supplier Risk
3. Customer Demographic and Location Coverage
4. Shipping, Returns and Service Risk
5. Popularity, Seasonality and Planning

3. SQL Concept Coverage

Concept	Number of queries
SELECT	7
WHERE	18
GROUP BY	25
CTE	10
VIEW	5
Window Function	10

4. Query Outputs and Interpretations

Q01. What are the first 10 product records?

Perspective: Affordability and Pricing | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT product_id, product_name, category, price, discount FROM diversified_ecommerce LIMIT 10;
```

Output sample:

product_id	product_name	category	price	discount
P6879	Jacket	Apparel	53.85	5
P5132	Camera	Electronics	761.26	10
P2941	Sneakers	Footwear	1756.76	5
P8545	Cookbooks	Books	295.24	10

P4594	Camera	Electronics	832.0	10

Interpretation: Shows a quick sample of products.

Q02. Which products are priced below 50?

Perspective: Affordability and Pricing | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, price FROM diversified_ecommerce WHERE price < 50 ORDER BY price LIMIT 20;
```

Output sample:

product_id	product_name	category	price
P7146	Slippers	Footwear	10.31
P4525	Toaster	Home Appliances	11.15
P4203	Blender	Home Appliances	11.41
P5334	Monitor	Electronics	11.57
P5051	Smartphone	Electronics	12.27

Interpretation: Finds lower-cost products.

Q03. Which products have high discounts?

Perspective: Affordability and Pricing | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, price, discount FROM diversified_ecommerce WHERE discount >= 20 ORDER BY discount DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	price	discount
P1241	Fiction	Books	1586.65	25
P2019	Hiking Shoes	Footwear	302.74	25
P2766	Boots	Footwear	1961.39	25
P5806	Sandals	Footwear	354.7	25
P8286	Gaming Console	Electronics	181.56	25

Interpretation: Identifies strongly discounted products.

Q04. Which product categories exist?

Perspective: Affordability and Pricing | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT category FROM diversified_ecommerce ORDER BY category;
```

Output sample:

category
Apparel
Books
Electronics
Footwear
Home Appliances

Interpretation: Lists categories.

Q05. Which products have final price above 1000?

Perspective: Affordability and Pricing | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, final_price_with_tax FROM diversified_ecommerce WHERE final_price_with_tax > 1000
ORDER BY final_price_with_tax DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	final_price_with_tax
P6660	Gaming Console	Electronics	2281.9795
P2938	Shorts	Apparel	2275.6315
P7345	Hiking Shoes	Footwear	2269.272
P7659	Headphones	Electronics	2266.512
P8744	Laptop	Electronics	2237.3232000000003

Interpretation: Shows expensive final-price products.

Q06. Which categories have the highest average final price?

Perspective: Affordability and Pricing | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(final_price_with_tax),2) AS avg_final_price, COUNT(*) AS products FROM diversified_ecommerce GROUP
BY category ORDER BY avg_final_price DESC;
```

Output sample:

category	avg_final_price	products
Footwear	1004.56	1008
Electronics	1001.64	1051
Home Appliances	984.1	965
Apparel	967.46	981
Books	966.77	995

Interpretation: Compares price levels by category.

Q07. Which categories have the most affordable products?

Perspective: Affordability and Pricing | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, COUNT(*) AS affordable_products FROM diversified_ecommerce WHERE final_price_with_tax < 100 GROUP BY category
ORDER BY affordable_products DESC;
```

Output sample:

category	affordable_products
Books	57
Home Appliances	54
Electronics	46
Apparel	42
Footwear	41

Interpretation: Counts affordable items by category.

Q08. What is average discount by category?

Perspective: Affordability and Pricing | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(discount),2) AS avg_discount, COUNT(*) AS products FROM diversified_ecommerce GROUP BY category
ORDER BY avg_discount DESC;
```

Output sample:

category	avg_discount	products
Footwear	12.82	1008
Apparel	12.66	981

Home Appliances	12.3	965
Electronics	12.29	1051
Books	12.27	995

Interpretation: Measures discounting patterns.

Q09. Which customer age groups see highest average product price?

Perspective: Affordability and Pricing | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_age_group, ROUND(AVG(final_price_with_tax),2) AS avg_final_price, COUNT(*) AS records FROM
diversified_ecommerce GROUP BY customer_age_group ORDER BY avg_final_price DESC;
```

Output sample:

customer_age_group	avg_final_price	records
45-54	1004.67	1033
55+	994.04	985
18-24	987.85	992
25-34	987.11	956
35-44	953.02	1034

Interpretation: Compares prices linked to age groups.

Q10. Which categories have high tax burden?

Perspective: Affordability and Pricing | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(tax_rate),2) AS avg_tax_rate, ROUND(AVG(final_price_with_tax - discounted_price),2) AS
avg_tax_value FROM diversified_ecommerce GROUP BY category ORDER BY avg_tax_rate DESC;
```

Output sample:

category	avg_tax_rate	avg_tax_value
Books	10.14	89.58
Footwear	10.04	91.58
Apparel	10.0	87.98
Home Appliances	9.98	89.31
Electronics	9.9	90.54

Interpretation: Looks at tax rate and tax value by category.

Q11. Which categories are above the overall average final price?

Perspective: Affordability and Pricing | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH cat AS (SELECT category, AVG(final_price_with_tax) AS avg_price FROM diversified_ecommerce GROUP BY category), overall AS
(SELECT AVG(final_price_with_tax) AS overall_avg FROM diversified_ecommerce) SELECT category, ROUND(avg_price,2) AS avg_price,
ROUND(overall_avg,2) AS overall_avg FROM cat, overall WHERE avg_price > overall_avg ORDER BY avg_price DESC;
```

Output sample:

category	avg_price	overall_avg
Footwear	1004.56	985.2
Electronics	1001.64	985.2

Interpretation: Compares category average price to overall average.

Q12. Rank categories by average final price.

Perspective: Affordability and Pricing | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT category, ROUND(AVG(final_price_with_tax),2) AS avg_final_price, RANK() OVER (ORDER BY AVG(final_price_with_tax) DESC)
AS price_rank FROM diversified_ecommerce GROUP BY category ORDER BY price_rank;
```

Output sample:

category	avg_final_price	price_rank
Footwear	1004.56	1
Electronics	1001.64	2
Home Appliances	984.1	3
Apparel	967.46	4
Books	966.77	5

Interpretation: Ranks categories by price.

Q13. Rank products by price within category.

Perspective: Affordability and Pricing | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT category, product_id, product_name, ROUND(final_price_with_tax,2) AS final_price, ROW_NUMBER() OVER (PARTITION BY
category ORDER BY final_price_with_tax DESC) AS price_rank_in_category FROM diversified_ecommerce ORDER BY category,
price_rank_in_category LIMIT 50;
```

Output sample:

category	product_id	product_name	final_price	price_rank_in_category
Apparel	P2938	Shorts	2275.63	1
Apparel	P6100	Jeans	2177.18	2
Apparel	P7454	Socks	2171.14	3
Apparel	P6246	Shirt	2167.57	4
Apparel	P8669	Socks	2114.1	5

Interpretation: Ranks products inside each category.

Q14. Use pricing view to show category pricing.

Perspective: Affordability and Pricing | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT category, ROUND(avg_final_price,2) AS avg_final_price, products FROM v_category_pricing ORDER BY avg_final_price DESC;
```

Output sample:

category	avg_final_price	products
Footwear	1004.56	1008
Electronics	1001.64	1051
Home Appliances	984.1	965
Apparel	967.46	981
Books	966.77	995

Interpretation: Uses reusable pricing view.

Q15. Which expensive products also have low discounts?

Perspective: Affordability and Pricing | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH thresholds AS (SELECT AVG(final_price_with_tax) AS avg_price, AVG(discount) AS avg_discount FROM diversified_ecommerce)
SELECT product_id, product_name, category, ROUND(final_price_with_tax,2) AS final_price, discount FROM diversified_ecommerce,
thresholds WHERE final_price_with_tax > avg_price AND discount < avg_discount ORDER BY final_price DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	final_price	discount
P6660	Gaming Console	Electronics	2281.98	0

P2938	Shorts	Apparel	2275.63	0
P7345	Hiking Shoes	Footwear	2269.27	0
P7659	Headphones	Electronics	2266.51	0
P8744	Laptop	Electronics	2237.32	0

Interpretation: Finds high-cost products with below-average discount.

Q16. Which products have low stock?

Perspective: Inventory and Supplier Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, stock_level FROM diversified_ecommerce WHERE stock_level < 50 ORDER BY stock_level
LIMIT 20;
```

Output sample:

product_id	product_name	category	stock_level
P3816	Smartwatch	Electronics	0
P7741	Cookbooks	Books	0
P6690	Blender	Home Appliances	0
P8143	Sweater	Apparel	0
P9081	Running Shoes	Footwear	0

Interpretation: Shows low-stock products.

Q17. Which products have very high stock?

Perspective: Inventory and Supplier Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, stock_level FROM diversified_ecommerce WHERE stock_level > 400 ORDER BY stock_level
DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	stock_level
P1093	Dishwasher	Home Appliances	500
P1539	Tablet	Electronics	500
P9869	Cookbooks	Books	500
P1732	Smartwatch	Electronics	500
P6397	Sweater	Apparel	500

Interpretation: Shows high-stock products.

Q18. Which suppliers appear in the data?

Perspective: Inventory and Supplier Risk | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT supplier_id FROM diversified_ecommerce ORDER BY supplier_id LIMIT 50;
```

Output sample:

supplier_id
S100
S101
S102
S103
S104

Interpretation: Lists suppliers.

Q19. Which suppliers provide electronics products?

Perspective: Inventory and Supplier Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT supplier_id, product_id, product_name, category FROM diversified_ecommerce WHERE category = 'Electronics' LIMIT 20;
```

Output sample:

supplier_id	product_id	product_name	category
S583	P5132	Camera	Electronics
S331	P4594	Camera	Electronics
S640	P8707	Smartwatch	Electronics
S689	P9427	Monitor	Electronics
S318	P1543	Smartphone	Electronics

Interpretation: Filters suppliers for one category.

Q20. Which records have low stock and high popularity?

Perspective: Inventory and Supplier Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, stock_level, popularity_index FROM diversified_ecommerce WHERE stock_level < 50 AND
popularity_index > 70 ORDER BY popularity_index DESC LIMIT 20;
```

Output sample:

product_id	product_name	stock_level	popularity_index
P8246	Formal Shoes	46	100
P1870	Speaker	10	100
P8979	Washing Machine	35	100
P9823	Slippers	38	100
P1877	Fiction	32	99

Interpretation: Finds potential replenishment priorities.

Q21. Which suppliers manage the most products?

Perspective: Inventory and Supplier Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT supplier_id, COUNT(*) AS products FROM diversified_ecommerce GROUP BY supplier_id ORDER BY products DESC LIMIT 20;
```

Output sample:

supplier_id	products
S377	15
S896	14
S736	13
S587	13
S231	13

Interpretation: Counts products by supplier.

Q22. Which categories have the lowest average stock?

Perspective: Inventory and Supplier Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(stock_level),2) AS avg_stock, COUNT(*) AS products FROM diversified_ecommerce GROUP BY category
ORDER BY avg_stock ASC;
```

Output sample:

category	avg_stock	products
Apparel	247.94	981
Home Appliances	247.95	965

Footwear	250.04	1008
Books	255.54	995
Electronics	258.35	1051

Interpretation: Compares stock levels by category.

Q23. Which suppliers have the highest stock value?

Perspective: Inventory and Supplier Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT supplier_id, ROUND(SUM(estimated_stock_value),2) AS stock_value, COUNT(*) AS products FROM diversified_ecommerce GROUP BY supplier_id ORDER BY stock_value DESC LIMIT 20;
```

Output sample:

supplier_id	stock_value	products
S587	4472974.74	13
S557	4078412.17	11
S231	4027864.37	13
S406	3981731.18	11
S584	3973250.73	12

Interpretation: Shows supplier stock value concentration.

Q24. Which categories have the most low-stock products?

Perspective: Inventory and Supplier Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, SUM(low_stock_flag) AS low_stock_products, COUNT(*) AS products FROM diversified_ecommerce GROUP BY category ORDER BY low_stock_products DESC;
```

Output sample:

category	low_stock_products	products
Footwear	105	1008
Home Appliances	97	965
Apparel	95	981
Electronics	91	1051
Books	88	995

Interpretation: Counts low-stock risk by category.

Q25. Which supplier-category combinations have high average popularity?

Perspective: Inventory and Supplier Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT supplier_id, category, ROUND(AVG(popularity_index),2) AS avg_popularity, COUNT(*) AS products FROM diversified_ecommerce GROUP BY supplier_id, category HAVING products >= 2 ORDER BY avg_popularity DESC LIMIT 20;
```

Output sample:

supplier_id	category	avg_popularity	products
S960	Home Appliances	99.0	2
S274	Apparel	98.0	2
S441	Books	98.0	2
S377	Footwear	97.0	2
S407	Home Appliances	97.0	2

Interpretation: Finds strong supplier-category combinations.

Q26. Rank suppliers by stock value.

Perspective: Inventory and Supplier Risk | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT supplier_id, ROUND(SUM(estimated_stock_value),2) AS stock_value, RANK() OVER (ORDER BY SUM(estimated_stock_value) DESC)
AS supplier_rank FROM diversified_ecommerce GROUP BY supplier_id ORDER BY supplier_rank LIMIT 20;
```

Output sample:

supplier_id	stock_value	supplier_rank
S587	4472974.74	1
S557	4078412.17	2
S231	4027864.37	3
S406	3981731.18	4
S584	3973250.73	5

Interpretation: Ranks suppliers by inventory value.

Q27. Which suppliers have above-average low-stock counts?

Perspective: Inventory and Supplier Risk | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH s AS (SELECT supplier_id, SUM(low_stock_flag) AS low_stock_count FROM diversified_ecommerce GROUP BY supplier_id), a AS
(SELECT AVG(low_stock_count) AS avg_low_stock FROM s) SELECT supplier_id, low_stock_count, ROUND(avg_low_stock,2) AS
avg_low_stock FROM s, a WHERE low_stock_count > avg_low_stock ORDER BY low_stock_count DESC LIMIT 20;
```

Output sample:

supplier_id	low_stock_count	avg_low_stock
S407	4	0.53
S104	3	0.53
S200	3	0.53
S223	3	0.53
S441	3	0.53

Interpretation: Uses CTEs to flag supplier low-stock risk.

Q28. Rank products by stock level within category.

Perspective: Inventory and Supplier Risk | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT category, product_id, product_name, stock_level, ROW_NUMBER() OVER (PARTITION BY category ORDER BY stock_level ASC) AS
low_stock_rank FROM diversified_ecommerce ORDER BY category, low_stock_rank LIMIT 50;
```

Output sample:

category	product_id	product_name	stock_level	low_stock_rank
Apparel	P8143	Sweater	0	1
Apparel	P1147	Dress	0	2
Apparel	P8749	Jeans	0	3
Apparel	P1759	Jacket	1	4
Apparel	P2182	Jacket	1	5

Interpretation: Ranks low-stock items within each category.

Q29. Use supplier risk view to show supplier inventory risk.

Perspective: Inventory and Supplier Risk | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT supplier_id, products, low_stock_products, ROUND(stock_value,2) AS stock_value FROM v_supplier_risk ORDER BY
low_stock_products DESC, stock_value DESC LIMIT 20;
```

Output sample:

supplier_id	products	low_stock_products	stock_value
S407	10	4	1799109.61
S441	7	3	2155471.83
S452	11	3	2105976.11
S104	9	3	1964862.49
S223	8	3	1763424.78

Interpretation: Uses reusable supplier risk view.

Q30. Which categories combine high popularity and low stock?

Perspective: Inventory and Supplier Risk | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH cat AS (SELECT category, AVG(popularity_index) AS avg_pop, AVG(stock_level) AS avg_stock FROM diversified_ecommerce GROUP BY category) SELECT category, ROUND(avg_pop,2) AS avg_popularity, ROUND(avg_stock,2) AS avg_stock FROM cat WHERE avg_pop > (SELECT AVG(avg_pop) FROM cat) AND avg_stock < (SELECT AVG(avg_stock) FROM cat) ORDER BY avg_popularity DESC;
```

Output sample:

category	avg_popularity	avg_stock

Interpretation: Identifies categories that may need replenishment.

Q31. Which customer locations appear?

Perspective: Customer Demographic and Location Coverage | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT customer_location FROM diversified_ecommerce ORDER BY customer_location LIMIT 50;
```

Output sample:

customer_location
Berlin, Germany
Cape Town, South Africa
Chicago, USA
Dubai, UAE
Houston, USA

Interpretation: Lists customer locations.

Q32. Which records are linked to UK customers?

Perspective: Customer Demographic and Location Coverage | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, category, customer_location, customer_age_group, customer_gender FROM diversified_ecommerce WHERE customer_location LIKE '%UK%' LIMIT 20;
```

Output sample:

product_id	category	customer_location	customer_age_group	customer_gender
P5132	Electronics	London, UK	25-34	Female
P6094	Books	London, UK	45-54	Female
P5172	Home Appliances	London, UK	25-34	Female
P1860	Home Appliances	London, UK	35-44	Female
P9313	Apparel	London, UK	35-44	Female

Interpretation: Filters UK-linked records.

Q33. Which records relate to age group 25-34?

Perspective: Customer Demographic and Location Coverage | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, category, customer_age_group, customer_gender FROM diversified_ecommerce WHERE customer_age_group = '25-34'
LIMIT 20;
```

Output sample:

product_id	category	customer_age_group	customer_gender
P5132	Electronics	25-34	Female
P2941	Footwear	25-34	Non-Binary
P5172	Home Appliances	25-34	Female
P9511	Apparel	25-34	Non-Binary
P4874	Home Appliances	25-34	Female

Interpretation: Filters by age group.

Q34. Which products are linked to female customers?

Perspective: Customer Demographic and Location Coverage | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, customer_gender FROM diversified_ecommerce WHERE customer_gender = 'Female' LIMIT
20;
```

Output sample:

product_id	product_name	category	customer_gender
P5132	Camera	Electronics	Female
P8545	Cookbooks	Books	Female
P1388	Non-Fiction	Books	Female
P7313	Running Shoes	Footwear	Female
P9687	Heels	Footwear	Female

Interpretation: Filters by gender.

Q35. Which gender groups appear?

Perspective: Customer Demographic and Location Coverage | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT customer_gender FROM diversified_ecommerce ORDER BY customer_gender;
```

Output sample:

customer_gender
Female
Male
Non-Binary

Interpretation: Lists gender groups.

Q36. Which customer locations have the most product records?

Perspective: Customer Demographic and Location Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_location, COUNT(*) AS records FROM diversified_ecommerce GROUP BY customer_location ORDER BY records DESC
LIMIT 20;
```

Output sample:

customer_location	records
Singapore	367
New York, USA	366
Houston, USA	358
Tokyo, Japan	350
Phoenix, USA	345

Interpretation: Counts product records by customer location.

Q37. What is average final price by age group?

Perspective: Customer Demographic and Location Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_age_group, ROUND(AVG(final_price_with_tax),2) AS avg_final_price, COUNT(*) AS records FROM
diversified_ecommerce GROUP BY customer_age_group ORDER BY avg_final_price DESC;
```

Output sample:

customer_age_group	avg_final_price	records
45-54	1004.67	1033
55+	994.04	985
18-24	987.85	992
25-34	987.11	956
35-44	953.02	1034

Interpretation: Compares prices by age group.

Q38. Which gender groups have highest average popularity?

Perspective: Customer Demographic and Location Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_gender, ROUND(AVG(popularity_index),2) AS avg_popularity, COUNT(*) AS records FROM diversified_ecommerce GROUP
BY customer_gender ORDER BY avg_popularity DESC;
```

Output sample:

customer_gender	avg_popularity	records
Female	49.35	1669
Non-Binary	49.17	1589
Male	49.14	1742

Interpretation: Compares popularity by gender group.

Q39. Which age-category combinations are most common?

Perspective: Customer Demographic and Location Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_age_group, category, COUNT(*) AS records FROM diversified_ecommerce GROUP BY customer_age_group, category
ORDER BY records DESC LIMIT 20;
```

Output sample:

customer_age_group	category	records
35-44	Electronics	227
35-44	Apparel	218
18-24	Footwear	217
45-54	Footwear	215
55+	Electronics	213

Interpretation: Shows category presence by age group.

Q40. Which locations have the highest average shipping cost?

Perspective: Customer Demographic and Location Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_location, ROUND(AVG(shipping_cost),2) AS avg_shipping_cost, COUNT(*) AS records FROM diversified_ecommerce
GROUP BY customer_location ORDER BY avg_shipping_cost DESC LIMIT 20;
```

Output sample:

customer_location	avg_shipping_cost	records
Houston, USA	25.59	358
Paris, France	25.51	343
Cape Town, South Africa	25.46	318
Berlin, Germany	25.37	324
Dubai, UAE	25.22	322

Interpretation: Compares shipping cost by location.

Q41. Rank categories within each age group by popularity.

Perspective: Customer Demographic and Location Coverage | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH x AS (SELECT customer_age_group, category, AVG(popularity_index) AS avg_pop FROM diversified_ecommerce GROUP BY customer_age_group, category) SELECT customer_age_group, category, ROUND(avg_pop,2) AS avg_popularity, ROW_NUMBER() OVER (PARTITION BY customer_age_group ORDER BY avg_pop DESC) AS category_rank FROM x ORDER BY customer_age_group, category_rank;
```

Output sample:

customer_age_group	category	avg_popularity	category_rank
18-24	Apparel	51.6	1
18-24	Electronics	51.28	2
18-24	Footwear	49.42	3
18-24	Books	49.04	4
18-24	Home Appliances	45.6	5

Interpretation: Ranks categories by age group.

Q42. Which locations have above-average shipping cost?

Perspective: Customer Demographic and Location Coverage | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH loc AS (SELECT customer_location, AVG(shipping_cost) AS avg_ship FROM diversified_ecommerce GROUP BY customer_location), overall AS (SELECT AVG(shipping_cost) AS overall_ship FROM diversified_ecommerce) SELECT customer_location, ROUND(avg_ship,2) AS avg_shipping_cost, ROUND(overall_ship,2) AS overall_avg FROM loc, overall WHERE avg_ship > overall_ship ORDER BY avg_ship DESC;
```

Output sample:

customer_location	avg_shipping_cost	overall_avg
Houston, USA	25.59	24.75
Paris, France	25.51	24.75
Cape Town, South Africa	25.46	24.75
Berlin, Germany	25.37	24.75
Dubai, UAE	25.22	24.75

Interpretation: Uses CTEs to compare location shipping cost.

Q43. Rank locations by product record volume.

Perspective: Customer Demographic and Location Coverage | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT customer_location, COUNT(*) AS records, RANK() OVER (ORDER BY COUNT(*) DESC) AS location_rank FROM diversified_ecommerce GROUP BY customer_location ORDER BY location_rank LIMIT 20;
```

Output sample:

customer_location	records	location_rank
Singapore	367	1
New York, USA	366	2

Houston, USA	358	3
Tokyo, Japan	350	4
Phoenix, USA	345	5

Interpretation: Ranks locations by coverage.

Q44. Use demographic view to inspect customer group coverage.

Perspective: Customer Demographic and Location Coverage | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT customer_age_group, customer_gender, records, ROUND(avg_final_price,2) AS avg_final_price FROM v_demographic_coverage
ORDER BY records DESC LIMIT 20;
```

Output sample:

customer_age_group	customer_gender	records	avg_final_price
45-54	Female	367	1029.26
35-44	Male	363	940.62
18-24	Male	358	997.62
45-54	Male	356	993.7
35-44	Female	342	963.55

Interpretation: Uses reusable demographic coverage view.

Q45. Which demographic groups have high price and high return rate?

Perspective: Customer Demographic and Location Coverage | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH demo AS (SELECT customer_age_group, customer_gender, AVG(final_price_with_tax) AS avg_price, AVG(return_rate) AS
avg_return FROM diversified_ecommerce GROUP BY customer_age_group, customer_gender) SELECT customer_age_group,
customer_gender, ROUND(avg_price,2) AS avg_price, ROUND(avg_return,2) AS avg_return_rate FROM demo WHERE avg_price > (SELECT
AVG(avg_price) FROM demo) AND avg_return > (SELECT AVG(avg_return) FROM demo) ORDER BY avg_price DESC;
```

Output sample:

customer_age_group	customer_gender	avg_price	avg_return_rate
45-54	Female	1029.26	10.59
25-34	Female	1000.53	10.76
18-24	Male	997.62	10.48
45-54	Male	993.7	10.48
45-54	Non-Binary	988.16	10.93

Interpretation: Finds customer groups linked to both higher price and returns.

Q46. Which shipping methods appear?

Perspective: Shipping, Returns and Service Risk | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT shipping_method FROM diversified_ecommerce ORDER BY shipping_method;
```

Output sample:

shipping_method
Express
Overnight
Standard

Interpretation: Lists shipping methods.

Q47. Which products have high return rate?

Perspective: Shipping, Returns and Service Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, return_rate FROM diversified_ecommerce WHERE return_rate >= 10 ORDER BY return_rate DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	return_rate
P3676	Hiking Shoes	Footwear	19.99
P2916	Air Conditioner	Home Appliances	19.99
P9683	Graphic Novels	Books	19.99
P9548	Slippers	Footwear	19.99
P6748	Skirt	Apparel	19.98

Interpretation: Filters high return-rate products.

Q48. Which products use overnight shipping?

Perspective: Shipping, Returns and Service Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, shipping_method, shipping_cost FROM diversified_ecommerce WHERE shipping_method = 'Overnight' LIMIT 20;
```

Output sample:

product_id	product_name	category	shipping_method	shipping_cost
P5132	Camera	Electronics	Overnight	20.88
P4594	Camera	Electronics	Overnight	45.93
P7313	Running Shoes	Footwear	Overnight	35.91
P9687	Heels	Footwear	Overnight	5.25
P3700	Fiction	Books	Overnight	3.35

Interpretation: Filters overnight shipping records.

Q49. Which products have expensive shipping?

Perspective: Shipping, Returns and Service Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, shipping_method, shipping_cost FROM diversified_ecommerce WHERE shipping_cost > 40 ORDER BY shipping_cost DESC LIMIT 20;
```

Output sample:

product_id	product_name	shipping_method	shipping_cost
P3735	Biographies	Standard	49.99
P6328	Shorts	Overnight	49.99
P3631	Laptop	Overnight	49.99
P1153	Socks	Standard	49.98
P3704	Slippers	Overnight	49.98

Interpretation: Shows high shipping cost cases.

Q50. Which records combine high return and high shipping cost?

Perspective: Shipping, Returns and Service Risk | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, return_rate, shipping_cost FROM diversified_ecommerce WHERE return_rate >= 10 AND shipping_cost > 30 ORDER BY return_rate DESC LIMIT 20;
```

Output sample:

product_id	product_name	return_rate	shipping_cost

P9548	Slippers	19.99	41.09
P6748	Skirt	19.98	41.99
P7689	Refrigerator	19.94	43.92
P5475	Jacket	19.93	40.84
P1747	Magazines	19.91	37.44

Interpretation: Finds service-risk records.

Q51. Which shipping method has highest average cost?

Perspective: Shipping, Returns and Service Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT shipping_method, ROUND(AVG(shipping_cost),2) AS avg_shipping_cost, COUNT(*) AS records FROM diversified_ecommerce GROUP BY shipping_method ORDER BY avg_shipping_cost DESC;
```

Output sample:

shipping_method	avg_shipping_cost	records
Standard	24.93	1694
Overnight	24.9	1661
Express	24.39	1645

Interpretation: Compares shipping cost by method.

Q52. Which categories have highest average return rate?

Perspective: Shipping, Returns and Service Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(return_rate),2) AS avg_return_rate, COUNT(*) AS products FROM diversified_ecommerce GROUP BY category ORDER BY avg_return_rate DESC;
```

Output sample:

category	avg_return_rate	products
Apparel	10.72	981
Footwear	10.67	1008
Home Appliances	10.52	965
Books	10.34	995
Electronics	10.11	1051

Interpretation: Compares returns by category.

Q53. Which shipping methods have highest return rate?

Perspective: Shipping, Returns and Service Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT shipping_method, ROUND(AVG(return_rate),2) AS avg_return_rate, COUNT(*) AS records FROM diversified_ecommerce GROUP BY shipping_method ORDER BY avg_return_rate DESC;
```

Output sample:

shipping_method	avg_return_rate	records
Standard	10.53	1694
Overnight	10.44	1661
Express	10.44	1645

Interpretation: Compares returns by shipping method.

Q54. Which categories have most high-return products?

Perspective: Shipping, Returns and Service Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, SUM(high_return_flag) AS high_return_products, COUNT(*) AS products FROM diversified_ecommerce GROUP BY
category ORDER BY high_return_products DESC;
```

Output sample:

category	high_return_products	products
Footwear	541	1008
Apparel	535	981
Electronics	515	1051
Books	507	995
Home Appliances	499	965

Interpretation: Counts high-return products.

Q55. Which location-shipping combinations are most expensive?

Perspective: Shipping, Returns and Service Risk | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_location, shipping_method, ROUND(AVG(shipping_cost),2) AS avg_shipping_cost, COUNT(*) AS records FROM
diversified_ecommerce GROUP BY customer_location, shipping_method ORDER BY avg_shipping_cost DESC LIMIT 20;
```

Output sample:

customer_location	shipping_method	avg_shipping_cost	records
Paris, France	Overnight	27.48	115
Berlin, Germany	Express	27.04	100
Cape Town, South Africa	Standard	26.76	104
Cape Town, South Africa	Overnight	26.38	100
New York, USA	Express	26.03	125

Interpretation: Shows expensive shipping location/method combinations.

Q56. Rank categories by average return rate.

Perspective: Shipping, Returns and Service Risk | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT category, ROUND(AVG(return_rate),2) AS avg_return_rate, RANK() OVER (ORDER BY AVG(return_rate) DESC) AS return_rank
FROM diversified_ecommerce GROUP BY category ORDER BY return_rank;
```

Output sample:

category	avg_return_rate	return_rank
Apparel	10.72	1
Footwear	10.67	2
Home Appliances	10.52	3
Books	10.34	4
Electronics	10.11	5

Interpretation: Ranks categories by return risk.

Q57. Which shipping methods are above average cost and return rate?

Perspective: Shipping, Returns and Service Risk | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH ship AS (SELECT shipping_method, AVG(shipping_cost) AS avg_ship, AVG(return_rate) AS avg_return FROM
diversified_ecommerce GROUP BY shipping_method) SELECT shipping_method, ROUND(avg_ship,2) AS avg_shipping_cost,
ROUND(avg_return,2) AS avg_return_rate FROM ship WHERE avg_ship > (SELECT AVG(avg_ship) FROM ship) AND avg_return > (SELECT
AVG(avg_return) FROM ship);
```

Output sample:

shipping_method	avg_shipping_cost	avg_return_rate
Standard	24.93	10.53

Interpretation: Flags high cost and high return shipping methods.

Q58. Rank products by return rate within category.

Perspective: Shipping, Returns and Service Risk | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT category, product_id, product_name, return_rate, ROW_NUMBER() OVER (PARTITION BY category ORDER BY return_rate DESC) AS return_rank_in_category FROM diversified_ecommerce ORDER BY category, return_rank_in_category LIMIT 50;
```

Output sample:

category	product_id	product_name	return_rate	return_rank_in_category
Apparel	P6748	Skirt	19.98	1
Apparel	P2361	Skirt	19.93	2
Apparel	P5475	Jacket	19.93	3
Apparel	P7303	Shorts	19.91	4
Apparel	P6476	Jacket	19.89	5

Interpretation: Ranks products by return risk within category.

Q59. Use service risk view to identify risky groups.

Perspective: Shipping, Returns and Service Risk | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT category, shipping_method, products, ROUND(avg_return_rate,2) AS avg_return_rate, ROUND(avg_shipping_cost,2) AS avg_shipping_cost FROM v_service_risk ORDER BY avg_return_rate DESC, avg_shipping_cost DESC LIMIT 20;
```

Output sample:

category	shipping_method	products	avg_return_rate	avg_shipping_cost
Apparel	Standard	341	10.85	25.86
Footwear	Overnight	314	10.83	23.65
Home Appliances	Express	307	10.81	24.39
Apparel	Express	309	10.69	25.43
Footwear	Standard	336	10.65	24.68

Interpretation: Uses reusable service risk view.

Q60. Which categories combine high return rate and high stock value?

Perspective: Shipping, Returns and Service Risk | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH cat AS (SELECT category, AVG(return_rate) AS avg_return, SUM(estimated_stock_value) AS stock_value FROM diversified_ecommerce GROUP BY category) SELECT category, ROUND(avg_return,2) AS avg_return_rate, ROUND(stock_value,2) AS stock_value FROM cat WHERE avg_return > (SELECT AVG(avg_return) FROM cat) AND stock_value > (SELECT AVG(stock_value) FROM cat) ORDER BY stock_value DESC;
```

Output sample:

category	avg_return_rate	stock_value
Footwear	10.67	259375612.56

Interpretation: Finds high-value categories with return risk.

Q61. Which products are seasonal?

Perspective: Popularity, Seasonality and Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, seasonality FROM diversified_ecommerce WHERE seasonality = 'Yes' LIMIT 20;
```

Output sample:

product_id	product_name	category	seasonality
P6879	Jacket	Apparel	Yes
P7313	Running Shoes	Footwear	Yes
P1241	Fiction	Books	Yes
P9687	Heels	Footwear	Yes
P3045	Fiction	Books	Yes

Interpretation: Filters seasonal products.

Q62. Which products are highly popular?

Perspective: Popularity, Seasonality and Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, popularity_index FROM diversified_ecommerce WHERE popularity_index >= 80 ORDER BY popularity_index DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	popularity_index
P9853	Comics	Books	100
P1263	Vacuum Cleaner	Home Appliances	100
P7082	Magazines	Books	100
P4443	Sweater	Apparel	100
P6567	Refrigerator	Home Appliances	100

Interpretation: Finds highly popular products.

Q63. Which products are non-seasonal and popular?

Perspective: Popularity, Seasonality and Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, seasonality, popularity_index FROM diversified_ecommerce WHERE seasonality = 'No' AND popularity_index >= 70 ORDER BY popularity_index DESC LIMIT 20;
```

Output sample:

product_id	product_name	category	seasonality	popularity_index
P4443	Sweater	Apparel	No	100
P8246	Formal Shoes	Footwear	No	100
P7806	Dress	Apparel	No	100
P3787	Jeans	Apparel	No	100
P5813	Textbooks	Books	No	100

Interpretation: Shows stable high-popularity products.

Q64. Which seasonality values exist?

Perspective: Popularity, Seasonality and Planning | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT seasonality FROM diversified_ecommerce ORDER BY seasonality;
```

Output sample:

seasonality
No
Yes

Interpretation: Checks seasonality categories.

Q65. Which products have low popularity?

Perspective: Popularity, Seasonality and Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT product_id, product_name, category, popularity_index FROM diversified_ecommerce WHERE popularity_index < 30 ORDER BY popularity_index LIMIT 20;
```

Output sample:

product_id	product_name	category	popularity_index
P4228	T-shirt	Apparel	0
P9692	Sandals	Footwear	0
P8521	Dishwasher	Home Appliances	0
P6108	Heels	Footwear	0
P8579	Headphones	Electronics	0

Interpretation: Finds low popularity items.

Q66. Which categories have highest average popularity?

Perspective: Popularity, Seasonality and Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(popularity_index),2) AS avg_popularity, COUNT(*) AS products FROM diversified_ecommerce GROUP BY category ORDER BY avg_popularity DESC;
```

Output sample:

category	avg_popularity	products
Electronics	51.19	1051
Footwear	48.95	1008
Apparel	48.62	981
Books	48.61	995
Home Appliances	48.59	965

Interpretation: Compares category popularity.

Q67. How many seasonal products are in each category?

Perspective: Popularity, Seasonality and Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, SUM(CASE WHEN seasonality = 'Yes' THEN 1 ELSE 0 END) AS seasonal_products, COUNT(*) AS products FROM diversified_ecommerce GROUP BY category ORDER BY seasonal_products DESC;
```

Output sample:

category	seasonal_products	products
Books	523	995
Electronics	512	1051
Home Appliances	494	965
Footwear	485	1008
Apparel	482	981

Interpretation: Counts seasonal products by category.

Q68. Which age groups have highest average popularity?

Perspective: Popularity, Seasonality and Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_age_group, ROUND(AVG(popularity_index),2) AS avg_popularity, COUNT(*) AS records FROM diversified_ecommerce GROUP BY customer_age_group ORDER BY avg_popularity DESC;
```

Output sample:

customer_age_group	avg_popularity	records

35-44	50.77	1034
55+	50.25	985
18-24	49.4	992
45-54	48.68	1033
25-34	46.89	956

Interpretation: Compares popularity across age groups.

Q69. Which categories have high popularity and low stock?

Perspective: Popularity, Seasonality and Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT category, ROUND(AVG(popularity_index),2) AS avg_popularity, ROUND(AVG(stock_level),2) AS avg_stock FROM
diversified_ecommerce GROUP BY category HAVING avg_popularity > 60 AND avg_stock < 250 ORDER BY avg_popularity DESC;
```

Output sample:

category	avg_popularity	avg_stock

Interpretation: Identifies planning opportunities.

Q70. Which shipping methods are linked to popular products?

Perspective: Popularity, Seasonality and Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT shipping_method, ROUND(AVG(popularity_index),2) AS avg_popularity, COUNT(*) AS records FROM diversified_ecommerce GROUP
BY shipping_method ORDER BY avg_popularity DESC;
```

Output sample:

shipping_method	avg_popularity	records
Standard	49.89	1694
Overnight	49.35	1661
Express	48.4	1645

Interpretation: Compares shipping method popularity.

Q71. Rank categories by average popularity.

Perspective: Popularity, Seasonality and Planning | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT category, ROUND(AVG(popularity_index),2) AS avg_popularity, RANK() OVER (ORDER BY AVG(popularity_index) DESC) AS
popularity_rank FROM diversified_ecommerce GROUP BY category ORDER BY popularity_rank;
```

Output sample:

category	avg_popularity	popularity_rank
Electronics	51.19	1
Footwear	48.95	2
Apparel	48.62	3
Books	48.61	4
Home Appliances	48.59	5

Interpretation: Ranks categories by popularity.

Q72. Which categories are above average popularity but below average stock?

Perspective: Popularity, Seasonality and Planning | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH cat AS (SELECT category, AVG(popularity_index) AS avg_pop, AVG(stock_level) AS avg_stock FROM diversified_ecommerce GROUP
BY category) SELECT category, ROUND(avg_pop,2) AS avg_popularity, ROUND(avg_stock,2) AS avg_stock FROM cat WHERE avg_pop >
(SELECT AVG(avg_pop) FROM cat) AND avg_stock < (SELECT AVG(avg_stock) FROM cat) ORDER BY avg_pop DESC;
```

Output sample:

category	avg_popularity	avg_stock

Interpretation: Uses CTEs to identify replenishment candidates.

Q73. Rank products by popularity within each shipping method.

Perspective: Popularity, Seasonality and Planning | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT shipping_method, product_id, product_name, popularity_index, ROW_NUMBER() OVER (PARTITION BY shipping_method ORDER BY popularity_index DESC) AS popularity_rank_in_method FROM diversified_ecommerce ORDER BY shipping_method, popularity_rank_in_method LIMIT 50;
```

Output sample:

shipping_method	product_id	product_name	popularity_index	popularity_rank_in_method
Express	P4443	Sweater	100	1
Express	P6567	Refrigerator	100	2
Express	P7854	Tablet	100	3
Express	P3787	Jeans	100	4
Express	P6945	Sandals	100	5

Interpretation: Ranks products within shipping methods.

Q74. Use popularity planning view for top categories.

Perspective: Popularity, Seasonality and Planning | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT category, seasonal_products, ROUND(avg_popularity,2) AS avg_popularity, ROUND(avg_stock,2) AS avg_stock FROM v_popularity_planning ORDER BY avg_popularity DESC;
```

Output sample:

category	seasonal_products	avg_popularity	avg_stock
Electronics	512	51.19	258.35
Footwear	485	48.95	250.04
Apparel	482	48.62	247.94
Books	523	48.61	255.54
Home Appliances	494	48.59	247.95

Interpretation: Uses reusable popularity planning view.

Q75. Which seasonal categories have high popularity and high return rate?

Perspective: Popularity, Seasonality and Planning | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH cat AS (SELECT category, AVG(popularity_index) AS avg_pop, AVG(return_rate) AS avg_return, SUM(CASE WHEN seasonality='Yes' THEN 1 ELSE 0 END) AS seasonal_count FROM diversified_ecommerce GROUP BY category) SELECT category, seasonal_count, ROUND(avg_pop,2) AS avg_popularity, ROUND(avg_return,2) AS avg_return_rate FROM cat WHERE seasonal_count > 0 AND avg_pop > (SELECT AVG(avg_pop) FROM cat) AND avg_return > (SELECT AVG(avg_return) FROM cat) ORDER BY avg_pop DESC;
```

Output sample:

category	seasonal_count	avg_popularity	avg_return_rate

Interpretation: Identifies seasonal popular categories with return risk.

5. Closing Reflection

The biggest challenge was turning a large operational dataset into structured analytical questions. The most useful insight is that SQL can quickly highlight where revenue, product demand, customer access, returns, and data-quality risks are concentrated. The project uses clear comments and query-by-query explanations so the logic can be explained in the 1-to-1 review call.